Mechanical Properties and Additive Manufacturing of Alkali-Activated Lunar Regolith in Artificial Lunar Environments

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Abstract. Lunar base construction has become a cutting-edge and multi-disciplinary research issue worldwide. The development of sustainable lunar infrastructure is essential for future exploration and a major challenge for constructing on the Moon is the availability of suitable building materials. In this regard, the research on alkali-activated lunar regolith simulants (AALRS) is a significant breakthrough. In this paper, the mechanical properties and additive manufacturing of AALRS are investigated with reference to the lunar environments. The study has shown that the type of alkali, curing temperature and vacuum exposure have a prominent effect on the solidification behaviour of AALRS. Furthermore, the acceptable long-term properties of silicate-based AALRS were also revealed after undergoing several lunar diel temperature cycles. The scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS) analysis have indicated the presence of two solidification mechanisms, geopolymerization and dehydration, within AALRS, which have contributed to its microstructure stability. The development of an innovative 3D printer for AALRS is a crucial step towards the future construction of a lunar base. Several conceptualized printers that can extrude, compress, and deposit treated lunar regolith will be presented. Additive manufacturing of lunar regolith has the potential to reduce the cost and increase the efficiency of building on the Moon.

Keyword: sustainable lunar construction · alkali-activated material · mechanical property · additive manufacturing

1 Introduction

"Earth is the cradle of humanity, but one cannot remain in the cradle forever (NASA, 2010)." The lunar base has been proposed by many countries as a multi-disciplinary, cutting-edge and strategic research issue nowadays. However, it is crucial for sustainable construction techniques using in-situ resources to be developed due to the huge launching cost and lunar severe environments, including large temperature differences,

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high vacuum, intense radiation and low gravity (Ding et al., 2019). Recently, several probable lunar construction schemes have been proposed, such as sintered lunar regolith (Fateri et al., 2019), sulfur concrete (Werkheiser et al., 2015), polymer concrete (Oh et al., 2020) and lunar geopolymer. The use of lunar geopolymer, which is an alkali-activated material made up of an aluminosilicate network, has shown promising results in terms of adequate strength, durability, and moderate processing conditions (Naser, 2019). This material therefore attracts an extensive research interest in the domain of sustainable lunar construction.

Due to the scarcity of actual lunar soil, semblable volcanic ash has been widely utilized to study the basic performances of alkali-activated lunar regolith simulant (AALRS). Under certain conditions, both sodium hydroxide (Zhang et al., 2022) and sodium silicate (Xiong et al., 2022) have been demonstrated to be effective in preparing AALRS. Therein, the temperature was considered a prominent factor. Some studies found the highest lunar daytime temperature could facilitate adequate early strength within several hours (Zhang et al., 2022). However, the influences of lunar high and cryogenic temperature cycles are still uncertain (Xiong et al., 2022; Zhang et al., 2022). In addition, recent researches were mostly implemented in the terrestrial atmosphere, but it lacked valid evidence on the effects of high vacuum (Davis et al., 2017). With respect to the materials shaping, current studies were mostly based on casting or hot-pressing (Montes et al., 2015), which consume excessive amounts of water or energy. Thus, alkali-activated lunar regolith satisfying high strength, outstanding durability and sustainable construction needs to be further investigated.

Considering the existing limitations, this work is focused on both early and long-term mechanical properties of AALRS under artificial lunar environments and aims to clarify the hardening and durability mechanisms. Furthermore, a conceptual AALRS 3D printing method will be proposed as a viable and sustainable solution for constructing in-situ infrastructure on the Moon.

2 Materials and Experiments

The lunar regolith simulant (LRS) in our previous work (Geng et al., 2023) was herein used as the raw material, which was made from volcanic ash in Huinan Town, Jilin Province. Table 1 lists the X-ray fluorescence (XRF) results of LRS and two target Apollo-17 samples (Heiken et al., 1991). They also exhibit similar lithologic minerals, morphological shape and mean particle diameter of ~ 40 μm (Geng et al., 2023). Additionally, sodium hydroxide (SH) and sodium silicate (SSx, x is SiO_2/Na_2O molar ratio) were selected as the alkali activators.

Table 1. Chemical composition of LRS and real lunar soils.

Type	Chemical composition percentage (wt%)							
	SiO_2	Al_2O_3	FeO	CaO	MgO	TiO_2	Na_2O	K_2O
LRS	43.4	17.4	16.4	9.1	5.2	2.4	2.8	1.9
Apollo-70051	42.2	15.7	12.4	11.5	10.3	5.1	0.3	0.1
Apollo-71501	43.2	17.1	12.2	11.8	10.4	4.2	0.4	0.1

Notes: The weight of Fe_2O_3 in LRS is converted into FeO in accordance with real lunar soils

The activators were first prepared as 30 wt% solutions with deionized water. The AALRS mixture was then created by rapidly mixing 13 wt% of alkaline solution and 87 wt% of LRS. The mixture was then compressed into blocks under 9.5 MPa and demolded. The specimen was either strictly sealed with a hot-melted bag or directly exposed in a vacuum tank under ~ 10 Pa. The samples were then placed in a digital oven that followed the lunar diel temperature variation (maximum 85 °C) with the cryogenic temperature realized by liquid N_2 (− 196 °C) as a substitute. This allowed for an artificial lunar environment similar to the lunar surface at a latitude of 40° (Williams et al., 2017). After being stored for the required amount of time, the samples were retrieved and their 3-point bending and uniaxial compressive strength were measured at least 3 times. Broken pieces were observed by SEM and EDS for microstructure characterization.

3 Solidification Properties

The type of alkali, curing temperature and vacuum exposure all significantly influence AALRS solidification. As illustrated in Fig. 1, both SH- and SS-based AALRS hardly solidify under a mild temperature (34 °C) within 24 h, even being protected from vacuum. When the curing temperature rises to 82 °C, SH-based AALRS generates an acceptable 24-h strength of 3.36 MPa for bending and 13.7 MPa for compression, much superior to SS-based ones. However, SS-based AALRS exhibits better solidification properties in vacuum exposure, and SS2.0 contributes to higher early strength than SS1.0. The SS2.0-based AALRS can establish 4.61 MPa of bending strength and 15.5 MPa of compressive strength within 5 h. Taking the availability into account, SH-based AALRS requires a higher curing temperature and a longer time to harden, adverse to the demands for sustainable lunar construction. Another crucial disadvantage of SH-based AALRS is the deteriorated solidification in a vacuum in contrast with SS-based ones. Therefore, SS with the modulus ≥ 2 is preferred as the alkali activator hereinafter.

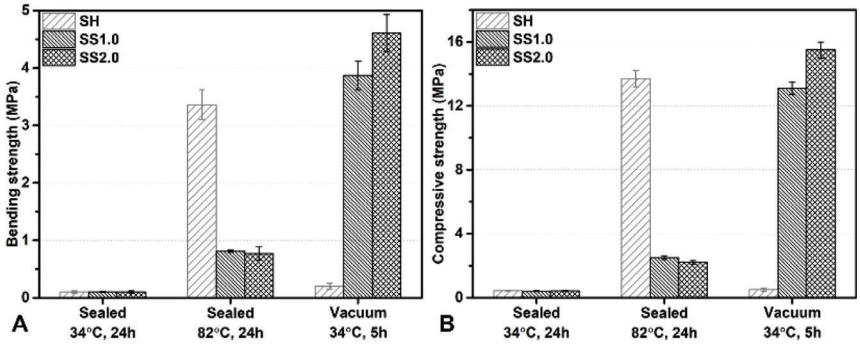

Fig. 1. The (A) bending and (B) compressive strength of AALRS at the solidification stage with different alkali types, curing temperatures and vacuum conditions.

4 Long-Term Properties

The long-term mechanical properties of AALRS undergoing certain lunar diel temperature cycles (~ 27 terrestrial days per cycle) are illustrated in Fig. 2. For all types of SS-based AALRS, their best mechanical properties are observed at the initial stage (solidified in vacuum for 5 h at 34 °C). After one lunar diel temperature cycle, the compressive strength of AALRS seems little variation, but the bending performance shows ~ 20% reduced strength with more brittleness. As the exposure of AALRS further prolongs to 4 cycles, all the groups still sustain tough bending and compressive strength with only a slight decrease within 5%. In general, this SS-based AALRS enables good durability under artificial lunar environments. It can be also found that SS2.5-based AALRS exhibits better bending and compressive strength than the other two SS during the whole service period, indicating a potential optimized modulus for SS to solidify lunar regolith.

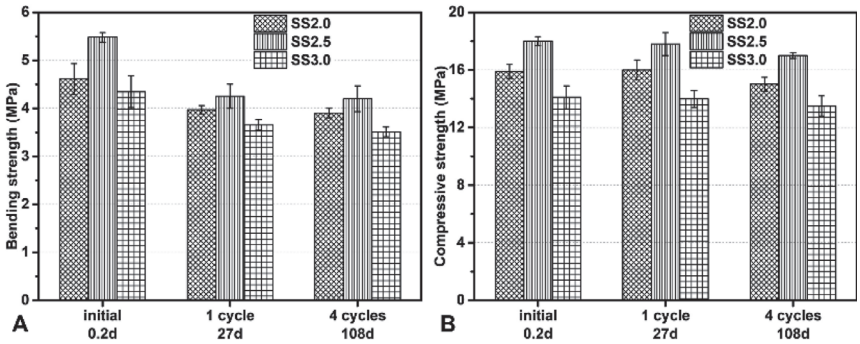

Fig. 2. The (A) bending and (B) compressive strength of AALRS at the initial solidification stage and after long-term lunar diel temperature cycles.

5 Microstructure Characterization

Figure 3 shows quite distinct microstructures of AALRS in different conditions. After being sealed at 82 °C for 24 h, granular products are found filling in the matrix of both types of AALRS. EDS indicates higher Na, Al and Si contents in these products compared

to the original LRS, implying the sodium aluminosilicate hydrates (N-A-S-H) generated from geopolymerization. However, the products change completely when being exposed in a vacuum for 5 h. Therein, unreacted alkali with dominant Na content is deposited as loose strips, resulting in the deteriorated strength of SH-based AALRS. However, regarding SS-based AALRS, smooth membranate products are observed connecting particles. This Si-rich film generated from SS dehydration promotes the early strength of SS-based AALRS in a vacuum.

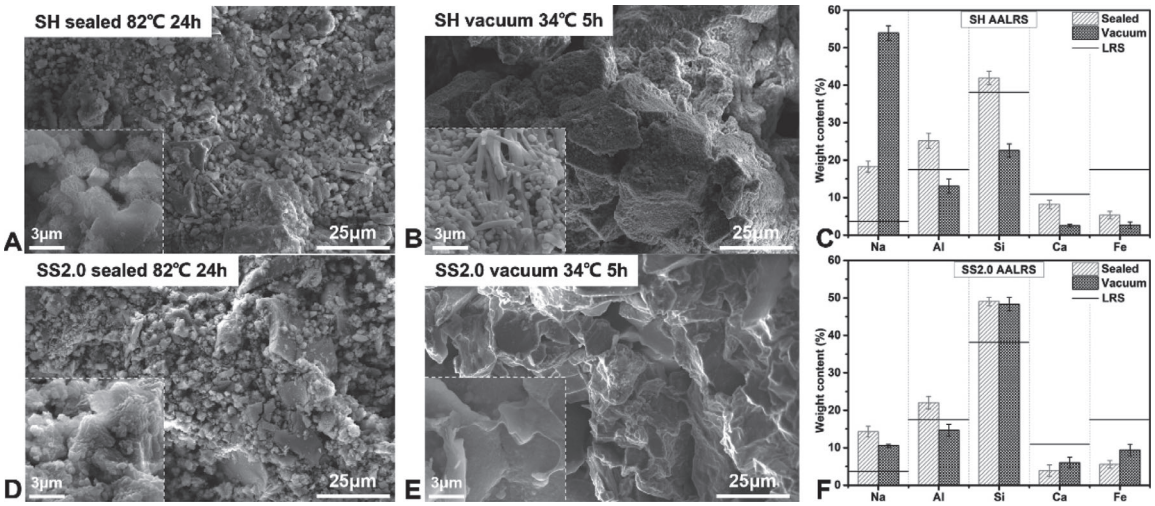

Fig. 3. (A-C) The microstructure and typical elements composition of SH-based AALRS. (D-F) The microstructure and typical elements composition of SS2.0-based AALRS.

The microstructure evolution of SS2.0-based AALRS exposed in the artificial lunar environments is illustrated in Fig. 4. During the 4 cycles, a relatively stable microstructure can be seen without obvious deterioration of the particle packing. The membranate binders are also found bonding LRS particles tightly after 4 cycles. This acceptable durability of SS-based AALRS might be attributed to the stability of dehydrated silicate under vacuum and large temperature difference.

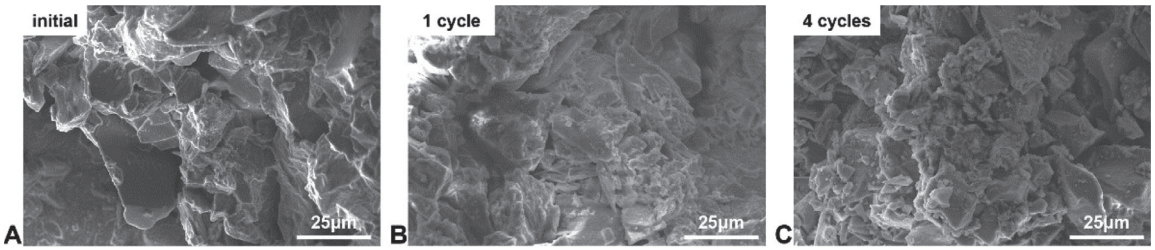

Fig. 4. The microstructure evolution of SS2.0-based AALRS at (A) initial stage and after (B) 1 lunar diel temperature cycle and (C) 4 lunar diel temperature cycles.

6 Future Prospects on Additive Manufacturing of Aalrs

Different from conventional cement-based concrete 3D printing, an innovative printing technique is required to match the unique material characteristics of AALRS. AALRS exhibits two solidification modes, namely geopolymerization and dehydration. While geopolymerization needs high temperature and long-term sealing, making it difficult for in-situ lunar additive manufacturing, dehydration is more feasible in lunar environments. To account for the limited availability of water and SS on the Moon, it is important to minimize their usage. As a result, the AALRS for 3D printing may be in the form of damp powdery ink. Extruding and compressing these damp powders can be challenging for lunar 3D printing.

A specific 3D printer is conceptualized in Fig. 5. The robotic arm is attached to a pipe system to transport treated lunar regolith to the desired position. The printhead is water-impermeable except for the material exit to reduce water evaporation before shaping. Damp lunar regolith is extruded and compressed through roller systems onto existing surfaces, and is expected to be dehydrated and rapidly hardened after leaving the printhead. This 3D printer is prospective to flexibly deposit lunar regolith in more places, including the ground, wall or roof, etc.

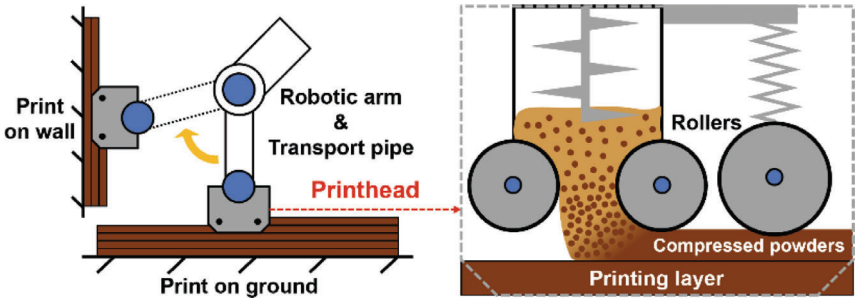

Fig. 5. The concept of a future lunar 3D printer for alkali-activated lunar regolith.

7 Conclusions

In this paper, a potential lunar construction material, alkali-activated lunar regolith, is investigated with respect to the mechanical properties and microstructures in artificial lunar environments. The effects of activator type, temperature and vacuum on AALRS solidification are revealed, indicating two hardening patterns, geopolymerization and dehydration. Based on the solidification mechanisms an optimized silicate activator is selected, and brings AALRS good durability after lunar diel temperature cycles and vacuum exposure. An innovative 3D printer design concept is also proposed for extruding, compressing and depositing treated lunar regolith according to the characteristics of AALRS, paving a prospective way to sustainably construct on the Moon.

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